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## **Ejercicios de Manejo de JSON**

### **1. Investigue cómo leer y escribir archivos JSON en Python aquí.**

#### **JSON basics in Python**

Python has a built-in module called json that lets you work with JSON data.

We start by importing it:

```
import json
```

#### ***Reading JSON (JSON → Python)***

There are two common ways:

1. From a JSON string in memory, using json.loads() (the “s” means string).

```
import json
```

```
json_text = '{"name": "Pikachu", "level": 25}'
```

```
data = json.loads(json_text) # data is now a Python dict
```

```
print(data["name"])      # Pikachu
```

2. From a JSON file on disk, using json.load().

```
import json
```

```
with open("pokemons.json", "r", encoding="utf-8") as f:
```

```
    pokemons = json.load(f) # pokemons is usually a list or dict
```

For my Pokémon exercise, I use json.load() to read the existing list of Pokémon from the file before adding a new one.

#### ***Writing JSON (Python → JSON)***

Again, there are two common tools:

1. To get a JSON string from a Python object, use json.dumps().

```
import json
```

```
pokemon = {"name": "Charmander", "level": 10}
```

```
json_text = json.dumps(pokemon) # JSON string
```

```
print(json_text)
```

2. To write JSON directly to a file, use json.dump().

```
import json
```

```
pokemons = [  
    {"name": "Pikachu", "level": 25},  
    {"name": "Charmander", "level": 10}  
]  
  
with open("pokemons.json", "w", encoding="utf-8") as f:  
    json.dump(pokemons, f, ensure_ascii=False, indent=4)
```

- ensure\_ascii=False keeps accents and non-ASCII characters readable.
- indent=4 makes the file nicely formatted for humans.

For example, for exercise 2: “**add a new Pokémon**” program, I will need to:

*Import json.*

Open the file and read the existing list with json.load().

Build a Python dict for the new Pokémon (with the fields your code expects).

Append that dict to the list.

And finally, I will open the file in write mode and save the updated list with json.dump()

\*\* See code attached to this assignment.